

Policy Learning with Surrogate Approximators for Block-Scale Urban Morphology Design: A Comparative Benchmarking Study

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Abstract

Integrating urban morphology generation, performance prediction, and design search is important for low-carbon urban design, yet direct simulation-based optimization remains computationally demanding at the block scale. This study develops and evaluates a surrogate-assisted deep reinforcement learning (DRL) workflow for urban morphology design using three energy-related indicators: Energy Use Intensity (EUI_t), Energy Generation (EG), and Sunlight Hours (H). A deep neural network (DNN) surrogate predicts these indicators from 12 morphological factors, and a Deep Deterministic Policy Gradient (DDPG) agent explores the design space under three reward preferences. The optimizer comparison is conducted on the most accurate surrogate identified from the dataset-scale study. Under this fair-budget benchmark, NSGA-II attains stronger objective summaries, the largest Hypervolume, and the lowest IGD, while all three DDPG scenarios remain weaker under post-hoc preference-aligned utility selection. The reward settings nevertheless still produce interpretable morphology-level differences, particularly in the shifts of openness, density, and solar-access descriptors across scenarios. Overall, the framework is better interpreted as a preference-conditioned region-finding and design-support tool than as a universally superior replacement for Pareto-based evolutionary search.

Keywords: Multi-objective optimization, Urban morphology, Energy performance, Deep reinforcement learning

Abbreviations and Symbols

Abbreviations

UM	Urban morphology
EP	Energy performance
UMF	Urban morphological factor
EPI	Energy performance indicator
DRL	Deep reinforcement learning
DNN	Deep neural network
DDPG	Deep Deterministic Policy Gradient
MOO	Multi-objective optimization
HV	Hypervolume
IGD	Inverted Generational Distance

Main symbols

EUI _t	Energy use intensity (kWh/m ² /y)
EG	Energy generation (10 ⁶ kWh/y)
H	Sunlight hours (h)
FAR	Floor area ratio
BD	Building density
OSR	Open space ratio
SVF	Sky view factor
AF	Average floors
AR _{e-w}	East-west aspect ratio
AR _{n-s}	North-south aspect ratio
OSLI	Open Space Location Index
θ	Block orientation angle (deg)
w_1, w_2, w_3	Reward weights for EUI _t , EG, and H
γ, τ	DDPG discount and soft-update factors

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1. Introduction

In the context of carbon peaking and carbon neutrality, the decarbonization of urban energy systems has become a central problem in sustainable development [1, 2]. Because cities concentrate both energy demand and renewable-energy opportunities, urban morphology (UM) strongly influences building energy performance (EP), solar access, and environmental quality [3, 4, 5]. In practice, however, urban design and energy planning are still often treated as separate tasks, which weakens block-scale energy performance and limits the effective use of renewable resources [6, 7]. Developing an integrated framework that links morphology generation, performance evaluation, and design optimization is therefore an important research problem at the intersection of architecture, urban planning, and energy systems [8].

This context also highlights the methodological importance of surrogate-based design studies. Large comparative sweeps over urban-energy design strategies are often bottlenecked by expensive simulation workflows, so surrogate-assisted optimization is increasingly used as a screening layer before full physical validation. Under these conditions, optimizer reliability becomes part of the energy-design problem itself: if a learning-based method is unstable, checkpoint-sensitive, or difficult to benchmark fairly, its practical value for urban energy design remains limited.

1.1. Research Motivation

To address this challenge, computational design methods based on parametric modeling and multi-objective optimization (MOO) have become mainstream [9, 10]. In this line of research, evolutionary algorithms such as NSGA-II are widely used to explore trade-offs among conflicting objectives including energy demand [11], photovoltaic potential [12], and daylight performance [13]. These studies have demonstrated the value of optimization-led urban design, but they also reinforce a common workflow in which the main analytical product is a Pareto archive. For design practice, however, a Pareto archive alone does not resolve whether a method can support preference-conditioned navigation, whether its search behaviour remains stable under repeated runs, or whether its benchmark advantage is robust to surrogate selection.

The core motivation of this study is therefore not to assume that policy learning should outperform Pareto search, but to test a more specific question: can a preference-conditioned policy-learning formulation identify useful high-performing regions of the same surrogate landscape while preserving interpretable morphology patterns under a shared optimization budget? Framed in this way, the study addresses both a design question and a methodological question, namely whether policy learning offers a useful decision-support bias for urban morphology design and how that bias should be interpreted relative to a fair NSGA-II benchmark.

1.2. Literature Review and Research Gap

Reviews of simulation-based optimization have established the methodological foundations of the field, but they also delimit the present gap. Nguyen et al. [14] and Machairas et al. [15] review building-performance optimization algorithms at a general level, yet they do not address how a policy-learning optimizer should be evaluated against a Pareto-search baseline in an urban morphology setting. More recent reviews by Pan et al. [18] and Westermann and Evins [20] clarify the computational value of simulation acceleration and surrogate modelling, but they stop short of asking how surrogate fidelity affects the credibility of optimizer ranking. Zhao et al. [19] further demonstrate coordinated surrogate-assisted optimization of building and microclimate performance, but their study does not examine whether the optimization method itself remains stable, preference-sensitive, and fairly benchmarked under a shared surrogate.

At the application level, existing urban-form optimization studies have demonstrated important capabilities, but each leaves part of the present problem unresolved. Ascione et al. [11] optimize envelope design, Li et al. [12] target photovoltaic-oriented inverse urban design, and Liu et al. [13] optimize urban block forms for daylight, energy use, and photovoltaic potential. These studies show that multi-objective search can recover high-performing trade-offs, but they still treat the main output as a set of Pareto solutions rather than as a preference-conditioned search process. Similarly, recent studies using NSGA-II or related evolutionary schemes in energy-oriented urban design [16, 17, 22, 23, 24, 25] provide stronger objective trade-offs, shading designs, or envelope-shape combinations, yet they do not establish whether designers can obtain a reusable policy for navigating the same design space, nor do they isolate how much of the reported advantage depends on the selected surrogate or benchmark protocol.

The reinforcement-learning literature does not close this gap either. General DRL surveys [26] and continuous-control benchmarks in robotics, games, language, and autonomous driving [27, 28, 29, 30] confirm that policy learning is effective in high-dimensional decision problems, while DQN [31] and DDPG [32] define the algorithmic basis for discrete and continuous control. However, these studies are not concerned with urban morphology design, surrogate-dependent benchmark fairness, or morphology-level interpretability. In the energy domain, DRL has been increasingly adopted for building HVAC control [33, 34] and power-system operation [35, 36], but these applications address operational control rather than static morphology generation. As a result, the literature still lacks a clear answer to a more specific scientific question: under a shared surrogate environment for block-scale urban morphology, can a preference-conditioned policy-learning formulation produce interpretable design patterns, and how does its empirical behaviour compare with a fair-budget NSGA-II benchmark?

This unresolved question defines the knowledge gap addressed in this paper. The gap is not the absence of optimization, surrogate modelling, or DRL in isolation; those elements already exist in the literature. Rather, the missing contribution is a benchmarked study that places a continuous-control DRL agent and NSGA-II in the same surrogate environment, evaluates them under matched budgets, and interprets the outcome not only through Pareto metrics but also through preference-conditioned design behaviour and morphology-level fingerprints.

1.3. Study Contributions and Paper Structure

Against this background, this study develops and evaluates a DRL-DNN integrated framework for block-scale urban morphology design. Three specialist agents with distinct reward preferences are trained and compared with an NSGA-II benchmark under the same surrogate environment.

1. This study formulates block-scale urban morphology optimization as a reward-weighted policy-learning task and embeds that task in a shared DNN surrogate environment for EUI_t, EG, and H.
2. This study evaluates whether different reward priorities lead to distinguishable search tendencies and morphology signatures within the same surrogate landscape, rather than treating optimization output only as a Pareto archive.
3. This study compares DDPG and NSGA-II under matched budgets in the same surrogate environment using Pareto-quality metrics and preference-aligned objective summaries, thereby addressing the benchmark-fairness gap identified above.
4. This study combines objective-space analysis with morphology-level interpretation, linear correlation analysis, and nonlinear response diagnostics to explain how key UMFs shape the observed performance patterns.

The remainder of this paper is organized as follows. Section 2 presents the DRL-DNN integrated framework, including parametric modeling, performance simulation, surrogate modeling, and the DDPG optimization engine. Section 3 reports the experimental results and comparative analysis. Section 4 discusses the implications and limitations of the study. Section 5 concludes the paper. Additional case-setting and surrogate-validation details are provided in Appendix A, and benchmark-robustness diagnostics are reported in Appendix B.

2. Methodology

This study proposes an integrated surrogate-assisted workflow for block-scale urban morphology design. As illustrated in Fig. 1, the framework combines parametric morphology generation, performance evaluation, surrogate modeling, and policy learning. A parametric dataset is first constructed to link urban morphological factors (UMFs) with key energy performance indicators (EPIs). A DNN surrogate is then trained for rapid performance prediction, after which a DDPG agent interacts with the surrogate environment to search for morphology strategies that balance energy demand, energy generation, and sunlight access.

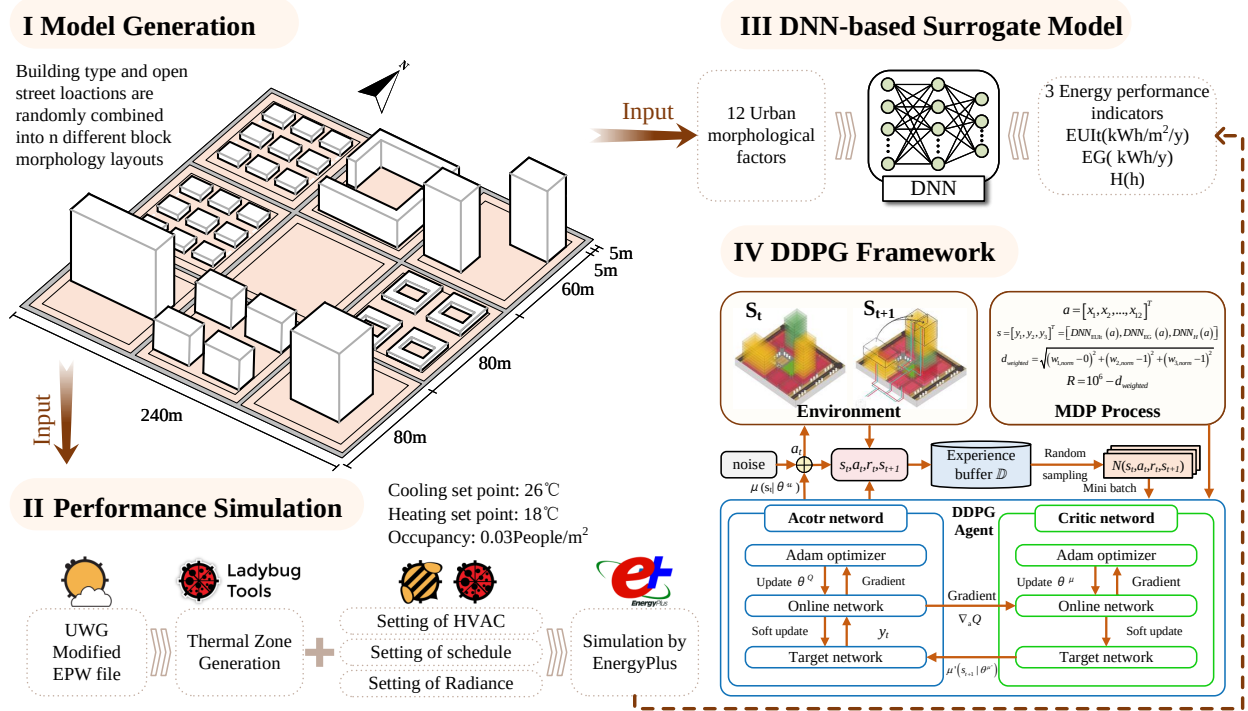


Figure 1: Overview of the proposed method. The left block summarizes parametric morphology generation and performance simulation, while the right block shows the surrogate-assisted optimization workflow used during policy learning.

2.1. Case Study and Parametric Modeling

The selected research area is Jianhu County, located in Jiangsu Province in the eastern coastal region of China. The county is situated between 33°16' and 33°41' N latitude and 119°33' and 120°05' E longitude, with an average elevation of 2 m. It has an administrative area of approximately 1160 km² and a population of around 600,000, resulting in a population density of about 520 persons/km². According to the Chinese Code for Thermal Design of Civil Buildings, Jianhu lies within the hot-summer and cold-winter climate zone and is classified as having a humid subtropical climate under the Koppen-Geiger system. The climate necessitates both building heating and cooling, as the average dry-bulb temperature is 27.1°C in the hottest month (July) and 1.6°C in the coldest month (January), with an annual average of 15.0°C. Furthermore, the region presents favorable conditions for solar energy utilization, receiving annual solar radiation of about 1250 kWh/m² and approximately 1800 hours of sunshine per year.

To generate a dataset that links urban form to energy performance, a parametric modeling approach was implemented on the Grasshopper platform. The core of this approach is a standard urban block subdivided into a 3 × 3 grid to create nine land units. Based on an analysis of the existing urban fabric, representative residential building prototypes, including point, slab, and courtyard types, were defined. In the parametric model, one land unit is designated as open space, while the remaining eight units are populated with these building prototypes to generate a wide variety of block morphologies.

This parametric generation process is controlled by twelve selected UMFs. These include Floor Area Ratio (FAR), Building Density (BD), Open Space Ratio (OSR), Average Floors (AF), Setback Diversity (SD), Shape Coefficient (SC), Average Perimeter-to-Area Ratio (PAR), the east-west and north-south aspect ratios (AR_{e-w} and AR_{n-s}), average Sky View Factor (SVF), Open Space Location Index (OSLI), and Block Orientation Angle θ . Once the geometric model of the block is configured on the Grasshopper platform, these UMFs can be calculated using its built-in tools. A dataset-scale study was carried out over 500, 1000, 1500, and 2000 morphologies, and the final optimizer comparison uses the most accurate surrogate identified from that study. The formulas used in this work are summarized in Table 1.

Table 1: Calculation formulas of the morphological factors.

Factor	Symbol	Formula
Floor area ratio	FAR	$\text{FAR} = \frac{\sum_{i=1}^n S_i}{A_s}$
Setback diversity	SD	$\text{SD} = h_{\max} - h_a$
Average floors	AF	$\text{AF} = \frac{\sum_{i=1}^n S_i}{\sum_{i=1}^n f_i}$
East-west aspect ratio	AR_{e-w}	$AR_{e-w} = \frac{1}{n} \sum_{i=1}^n \frac{H_i}{W_{i,e-w}}$
North-south aspect ratio	AR_{n-s}	$AR_{n-s} = \frac{1}{n} \sum_{i=1}^n \frac{H_i}{W_{i,n-s}}$
Sky view factor	SVF	$\text{SVF} = \frac{1}{2\pi} \int_0^{2\pi} \cos^2(\beta(R, \theta)) d\theta$
Building density	BD	$\text{BD} = \frac{\sum_{i=1}^n f_i}{A_s}$
Open space ratio	OSR	$\text{OSR} = \frac{A_s - \sum_{i=1}^n f_i}{\sum_{i=1}^n S_i}$
Shape coefficient	SC	$\text{SC} = \frac{\sum_{i=1}^n c_i}{\sum_{i=1}^n h_i f_i}$
Perimeter-to-area ratio	PAR	$\text{PAR} = \frac{\sum_{i=1}^n p_i}{\sum_{i=1}^n f_i}$

Note: n is the number of buildings; S_i is the total floor area of building i ; A_s is the block area; f_i is the building footprint area of building i ; $h_{\max}$ and h_a are the maximum and average building heights of the block; H_i is the building height; $W_{i,e-w}$ and $W_{i,n-s}$ are the east-west and north-south canyon widths associated with building i ; c_i is the surface area of building i ; p_i is the perimeter of building i ; and β is the angle from the center point to the maximum obstacle height at a distance equal to the constant search radius R .

In addition to the factors in Table 1, OSLI is introduced to characterize the topological position of the open space within the 3×3 grid and to account for its influence on local wind, thermal, and daylight conditions [17, 18]. OSLI is an integer variable ranging from 0 to 8. Furthermore, the block orientation angle θ , ranging from $[-45^\circ, 45^\circ]$, is included to regulate solar access. Additional geometric details of the block layout and the representative building prototypes are provided in Appendix A.

2.2. Performance Simulation and Optimization Objectives

This study evaluates the energy behavior and solar potential of residential blocks using three key EPIs: total Energy Use Intensity (EUIt), Energy Generation (EG), and Sunlight Hours (H). Together, these indicators represent the demand, supply, and daylight dimensions of block-scale energy performance.

The three targets retain the same physical interpretation as in the original Ladybug/Honeybee/EnergyPlus workflow, but the dataset used in the present study was generated with the documented fallback analytic simulator rather than direct full-stack simulation calls. In the synced dataset metadata, this is recorded as a fallback analytic simulation mode. The evidence in this study should therefore be interpreted as surrogate-assisted optimization on an analytically reconstructed response surface that preserves the intended demand-supply-daylight trade-offs within the sampled design domain.

Under this response definition, H denotes daylight access over the winter design window represented in the original workflow by south-facing ground-floor windowsill test points on a typical coldest day (January 20, 08:00–16:00). In the present dataset, the fallback simulator approximates this daylight response from the morphological factors while preserving the same directional interpretation: higher H indicates better solar access.

EUIt remains the energy-demand target of the framework and corresponds conceptually to annual residential energy use under the envelope, internal-load, and operating assumptions of the original Honeybee/EnergyPlus model. In the present dataset, the fallback simulator approximates this demand response from the same morphology descriptors rather than from direct annual EnergyPlus execution.

EG remains the renewable-generation target and corresponds conceptually to annual rooftop PV potential under the standardized assumptions of the original Ladybug-Radiance workflow. In the present dataset, the fallback simulator approximates this generation response from the morphology descriptors while preserving the same higher-is-better interpretation.

Table 2: Reference assumptions of the original physics-based building-energy workflow. The results reported in this manuscript are generated from the documented fallback analytic simulator rather than direct EnergyPlus runs.

Classification	Parameter	Specification
System settings	Weather data	EPW files modified based on UWG urban microclimate simulations
System settings	Simulation period	Annual simulation (January 1 to December 31)
Envelope thermal performance	Wall U-value	0.8 W/(m ² ·K)
Envelope thermal performance	Roof U-value	0.5 W/(m ² ·K)
Envelope thermal performance	Floor U-value	1.5 W/(m ² ·K)
Envelope thermal performance	Window properties	U-value = 2.70 W/(m ² ·K); Solar Heat Gain Coefficient (SHGC) = 0.78
Window-to-wall ratio	North & South facades	0.4
Window-to-wall ratio	East & West facades	0.1
Internal loads	Occupant density	0.03 persons/m ² ; metabolic rate = 100 W/person
Internal loads	Lighting power density	5 W/m ²
Internal loads	Equipment power density	1.9 W/m ²
Internal loads	Ventilation rate	30 m ³ /(h·person)
HVAC system	Thermostat setpoints	Heating: Dec 1–Feb 28, 18°C; Cooling: Jun 15–Aug 31, 26°C
HVAC system	Infiltration rate	1 ACH (natural air leakage)

2.3. DNN-based Surrogate Model

In this study, a DNN is constructed as a predictive surrogate model to estimate the three target performance indicators from the simulated dataset. The model adopts a multi-layer feedforward architecture in which information flows from the input layer through the hidden layers to the output layer. Through stacked fully connected transformations, the network captures the nonlinear relationships between urban morphological factors and block-scale performance outcomes.

The DNN training process consists of two key stages: forward propagation and backpropagation. In the forward stage, the network processes input data to generate a prediction. During backpropagation, the prediction error is propagated backward from the output layer to the input layer. This process involves calculating the gradient of the loss function with respect to the network weights and iteratively updating the model parameters to reduce prediction error:

$$\mathbf{w} = \mathbf{w} + \eta \nabla_{\mathbf{w}} = \mathbf{w} + \eta \text{backpropagation}(f(\mathbf{x})), \quad (1)$$

where η is the learning rate and $f(\mathbf{x})$ is the objective function to minimize.

To determine the DNN architecture and training settings, Bayesian Optimization (BO) was used for automated hyperparameter tuning. Compared with grid search, BO guides the search process by constructing a probabilistic model of the objective function and repeatedly selecting promising hyperparameter combinations. After the optimal settings were determined, model performance was evaluated using five-fold cross-validation. Because the surrogate is subsequently used for optimization, it is treated here as an optimization-enabling approximation rather than as a direct replacement for the original simulation workflow.

Table 3: Optimal hyperparameters of the DNN determined by Bayesian Optimization.

Hyperparameter	Value
Number of dense layers	1
Units of 1st layer	128
Units of 2nd layer	–
Dropout rate	0.143
Learning rate	0.00134
Batch size	32
Optimizer	Adam
Activation	ReLU

To ensure a rigorous evaluation of generalization capability and avoid bias from a single train-test split, the final

highest-accuracy surrogate was evaluated by five-fold cross-validation over the 2000-sample regime. The coefficient of determination (R^2) and Mean Absolute Error (MAE) were used as key evaluation metrics.

Table 4: Summary of DNN prediction performance using five-fold cross-validation.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUI _t	0.983	0.209	± 0.001	± 0.005
EG	0.972	0.026	± 0.002	± 0.001
H	0.992	0.025	± 0.001	± 0.001

The evaluation results show that the selected 2000-sample surrogate maintains R^2 values close to 1 for all three objectives while keeping MAE low relative to the observed target ranges. In the scale study, this model achieves a mean target nMAE of 0.0182 and a mean tail nMAE of 0.0203, which together indicate the strongest predictive accuracy among the tested candidates. Taken together, these results justify using the selected 2000-sample surrogate as the basis for the final optimizer comparison.

2.4. DDPG-based Optimization Framework

The parametric modeling process and the DNN surrogate together provide the basis for the integrated optimization framework linking urban morphology and energy performance. To optimize these indicators jointly, a DRL framework was developed, as shown in Fig. 2. The framework has two components. The first defines the optimization problem through the environment, action space, state representation, and reward. The second implements the DDPG agent, which interacts with the environment through an experience replay buffer and updates its policy iteratively.

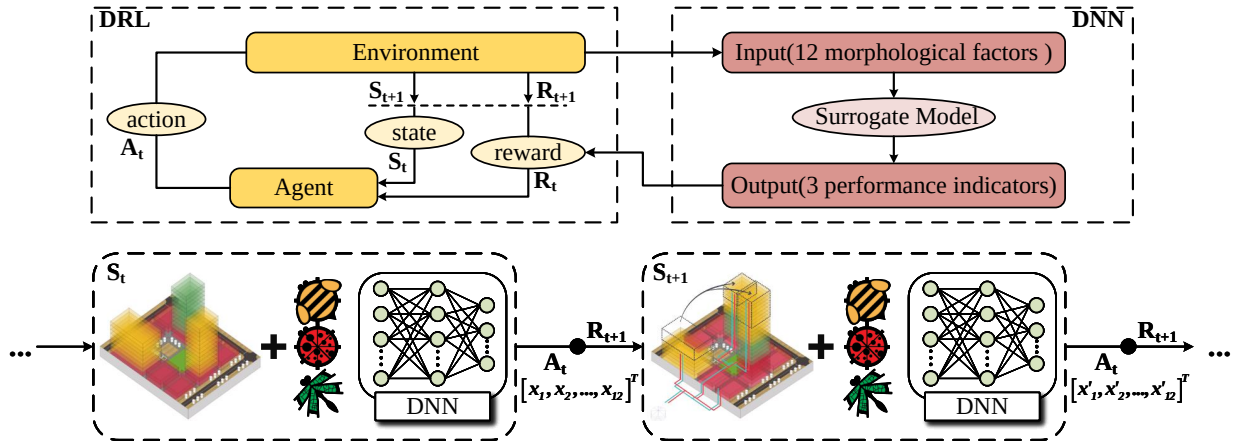


Figure 2: Conceptual framework of the DRL-based optimization. The upper part summarizes the relationship between DRL, the environment, and the DNN surrogate; the lower part details the stepwise Markov Decision Process used by the actor to query the surrogate and receive updated state and reward signals.

2.4.1. Problem Formulation

To address the MOO problem in urban energy performance, this study formulates the optimization process as a Markov Decision Process (MDP), typically defined by the four-tuple (S, A, P, R) . The DRL agent continuously receives state and reward signals from the environment and updates its policy iteratively to improve the morphology solution.

Action space. In the DRL framework, an action is the fundamental unit of interaction between the agent and the environment. In this study, an action is defined as an adjustment to a set of preset building design parameters. Specifically, the action vector is formed by twelve key design parameters:

$$\mathbf{a} = [x_1, x_2, \dots, x_{12}]^T, \quad (2)$$

where $\mathbf{a}$ represents a specific action selected by the agent and x_i is the selected value for the i th design parameter.

State space. The state is defined as the vector of the key EPIs predicted by the DNN-based surrogate model:

$$\mathbf{s} = [y_1, y_2, y_3]^T = [\text{DNN}_{\text{EUIt}}(\mathbf{a}), \text{DNN}_{\text{EG}}(\mathbf{a}), \text{DNN}_{\text{H}}(\mathbf{a})]^T, \quad (3)$$

where y_1 , y_2 , and y_3 are the predicted values of EUIt, EG, and H, respectively.

State transition probability. The state-transition probability $P(s' | s, a)$ defines the probability distribution of the environment transitioning to the next state given the current state and action. In the model-free DRL method employed here, P is not modeled explicitly; instead, the agent learns directly from samples obtained through interaction with the environment. In our framework, for a given action, the DNN surrogate deterministically or near-deterministically outputs the corresponding performance indicators, thereby defining the next state.

Reward function. The reward function provides the scalar feedback signal that guides learning. Because this study simultaneously minimizes EUIt while maximizing EG and H, a normalized weighted-distance formulation is adopted. First, the three objectives are mapped to a common $[0, 1]$ interval:

$$\begin{aligned} y_{1,\text{norm}} &= \frac{y_1 - y_{1,\min}}{y_{1,\max} - y_{1,\min}}, \\ y_{2,\text{norm}} &= \frac{y_2 - y_{2,\min}}{y_{2,\max} - y_{2,\min}}, \\ y_{3,\text{norm}} &= \frac{y_3 - y_{3,\min}}{y_{3,\max} - y_{3,\min}}. \end{aligned} \quad (4)$$

In the normalized space, the utopian point is defined as $[0, 1, 1]$, representing the ideal state in which all objectives simultaneously reach their best values. The weighted Euclidean distance between the current state and the utopian point is then calculated as

$$d_{\text{weighted}} = \sqrt{(w_1 y_{1,\text{norm}} - 0)^2 + (w_2 y_{2,\text{norm}} - 1)^2 + (w_3 y_{3,\text{norm}} - 1)^2}, \quad (5)$$

where w_1 , w_2 , and w_3 are preset non-negative weight coefficients whose sum equals 1. In this study, the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. This choice makes the reward scale dependent on dataset coverage, and therefore the bound selection must be interpreted as a practical normalization strategy rather than a theoretically unique definition. Finally, the reward function is defined as

$$R = 10^6 - d_{\text{weighted}}. \quad (6)$$

The reward function therefore transforms the multi-objective design problem into a scalar optimization target that still preserves design preferences through the weight coefficients. The factor 10^6 is used only as a numerical offset to keep rewards positive and to stabilize optimization logs; it does not alter the ranking induced by the weighted distance.

2.4.2. DDPG Agent Implementation

To solve the formulated MDP, this study adopts the DDPG algorithm. DDPG is a model-free actor-critic method well suited to high-dimensional continuous action spaces. It concurrently trains an actor network, which selects actions, and a critic network, which evaluates the value of those actions under the reward function. Here, the focus is on the implementation used for urban morphology optimization, while additional convergence and benchmark details are reported in Appendix B.

The implementation is structured around a multi-scenario analysis designed to investigate the agent's ability to generate targeted design solutions based on varying decision-maker preferences. Multiple independent DDPG agents are trained, each guided by a unique reward function reflecting a specific design priority. The common architectural hyperparameters are summarized in Table 5.

Table 5: Hyperparameter settings for the DDPG agent.

Parameter	Value
Reward discount factor (γ)	0.999
Learning rate (actor)	0.001
Learning rate (critic)	0.002
Layers (actor network)	2
Units per layer (actor)	64, 32
Layers (critic network)	2
Units per layer (critic)	64, 32
Soft update factor (τ)	0.001
Batch size	128
Replay buffer size	1×10^6
Maximum episodes	600
Max steps per episode	40
Noise decay factor	0.9998

The actor and critic networks both use two hidden layers with 64 and 32 neurons. This relatively simple architecture was found to be effective for the problem complexity while avoiding the instability often associated with deeper networks. The critic learning rate is slightly larger than the actor learning rate, allowing the value function to stabilize faster and provide a reliable training signal to the policy network. The discount factor is set to 0.999, encouraging farsighted strategies focused on long-term cumulative rewards.

To ensure adequate exploration, Gaussian noise with an initial variance of 1.0 is added to the actor outputs and gradually reduced using a decay factor of 0.9998. A replay buffer with a capacity of one million transitions is used to decorrelate samples and stabilize learning; at each training step, a mini-batch of 128 transitions is sampled to update the networks.

The central pillar of the implementation is the configuration of three distinct optimization scenarios. These scenarios are realized by adjusting the weight coefficients (w_1, w_2, w_3) in the reward function:

- **Scenario 1: Balanced performance.** Equal weights are assigned, $w_1 = w_2 = w_3 = \frac{1}{3}$, producing an unbiased preference among minimizing EUIt, maximizing EG, and maximizing H.
- **Scenario 2: Energy saving focus.** The weights are set to $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$, prioritizing reduced energy consumption.
- **Scenario 3: Energy generation focus.** The weights are set to $(w_1, w_2, w_3) = (0.2, 0.6, 0.2)$, prioritizing renewable energy production.

By training a separate agent for each of these scenarios, the DRL framework is transformed from a single-solution optimizer into a flexible decision-support tool that supports a priori articulation of design intent. The overall 1...404 tokens truncated...ons remain tightly concentrated around the one-to-one line for all three objectives, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. Even so, these error statistics should be interpreted together with the extreme-region diagnostics and candidate re-evaluation results, because optimization concentrates attention on the parts of the response surface where surrogate misspecification is most consequential.

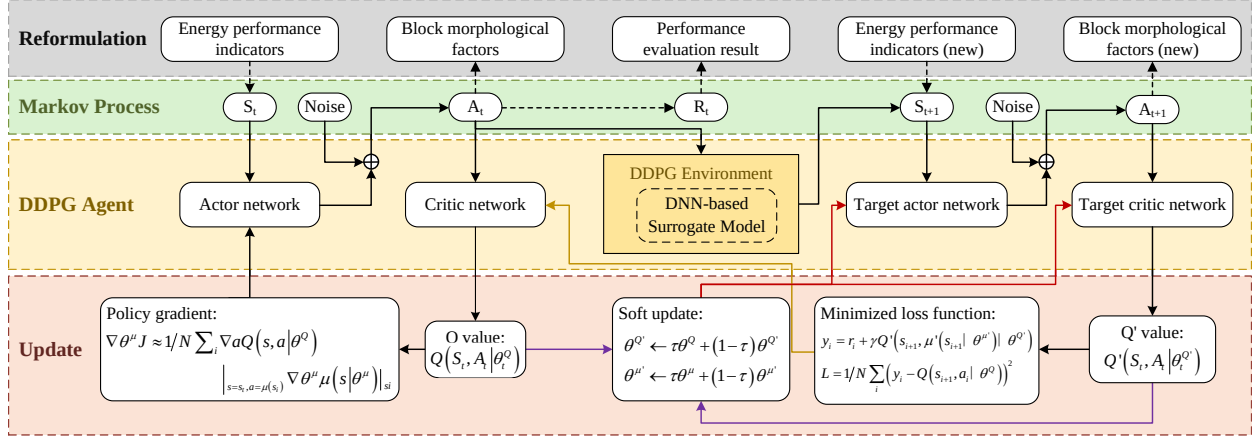


Figure 3: Architecture of the DDPG agent, showing the actor-critic structure and learning process.

3. Results

3.1. Surrogate Model Performance

Based on the simulations generated through the Grasshopper process, a surrogate scale study was carried out over 500, 1000, 1500, and 2000 simulated morphologies to identify the most accurate model available for this study. The final optimizer comparison uses the 2000-sample DNN with standard scaling on both inputs and targets and a single hidden layer with 128 units. To reduce overfitting and training instability during candidate selection, early stopping and dropout were employed throughout the search.

As summarized in Tables 3 and 4, the final DNN achieved high predictive accuracy and strong generalization on the expanded 2000-sample regime. Appendix A further reports RMSE values of 0.279 for EUIt, 0.033 for EG, and 0.032 for H, with normalized RMSE values below 3.1% for all three targets. The parity plots in Fig. 4 show that the predictions remain tightly concentrated around the one-to-one line, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. These statistics should nevertheless be interpreted together with the extreme-region diagnostics in Appendix A and the candidate re-evaluation results in Appendix B, because optimization is most sensitive to misspecification in the regions of the response surface that repeatedly attract the search process.

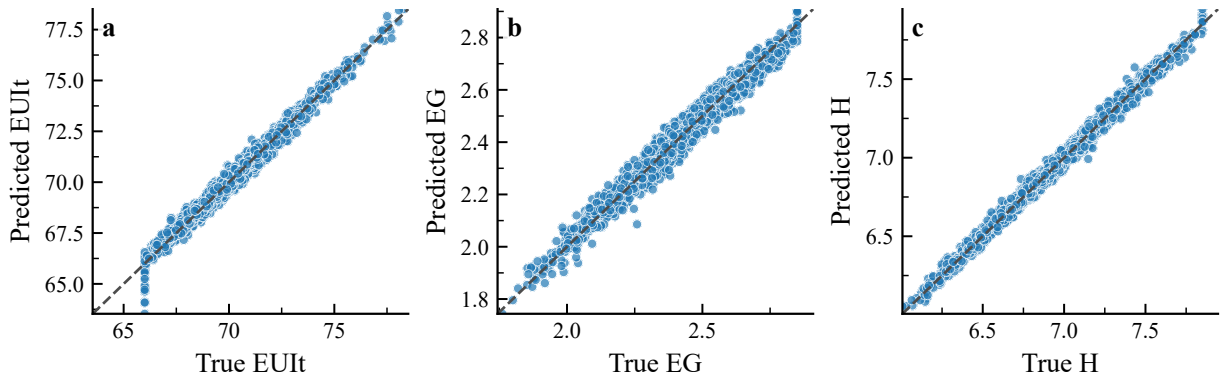


Figure 4: Parity plots comparing surrogate predictions against simulated targets for the three objectives: (a) EUIt, (b) EG, and (c) H. The dashed line denotes perfect agreement.

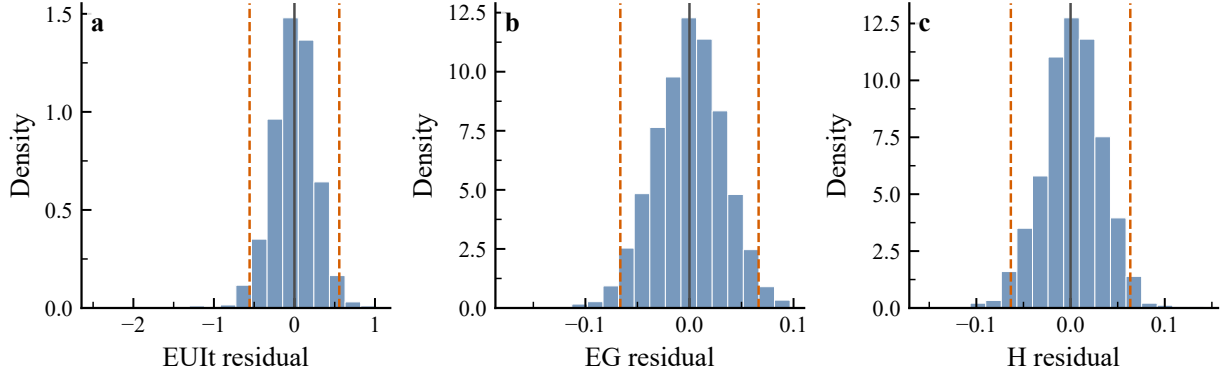


Figure 5: Residual distributions of the surrogate predictions for the three objectives: (a) EUIt, (b) EG, and (c) H. Dashed lines indicate the $\pm 2\sigma$ range around zero residual.

3.2. DDPG Training Dynamics

The learning dynamics of the agents were monitored to assess whether the policy updates approached a recurrent high-reward regime. Figure 6 summarizes the mean learning curves and one-standard-deviation envelopes across the available seeds for the three DDPG scenarios. Each scenario was trained with 20 seeds, and each seed was run for 600 episodes under the guarded surrogate evaluator, which provides a more informative view of training behavior than a single-curve presentation.

The multi-seed curves show that cumulative reward increases sharply during training and then transitions into a narrower high-reward regime, indicating a shift from broad exploration to more concentrated surrogate-region search. On the selected 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401, 320, and 398 for the balanced, saving-focused, and generation-focused scenarios, respectively. The balanced scenario is visibly smoother and exhibits lower cross-seed fluctuation in reward and output trajectories than the other two scenarios, whereas the saving-focused and generation-focused scenarios do not become uniformly less variable across all metrics. At the same time, late-stage regression remains common, with mean best-to-final reward gap ratios around 0.62, 0.71, and 0.71 and substantial-regression fractions of 95%, 85%, and 100% across the three scenarios. These curves should therefore be interpreted as evidence that policy learning can reach high-performing surrogate regions rather than as proof of stable asymptotic convergence or global optimality. The learning-curve evidence is therefore complemented by benchmark comparisons and independent candidate re-evaluation.

The three scenarios remain distinguishable but still overlap in aggregate objective space. In the 20-seed experiments on the selected surrogate, the energy-saving-focused scenario yields the lowest mean EUIt (67.10 kWh/m²/y), the generation-focused scenario yields the highest mean EG (2.763×10^6 kWh/y), and the balanced scenario retains the highest mean H (7.687 h). These differences should be interpreted as preference-conditioned shifts within one surrogate landscape rather than as proof of sharply separated optima.

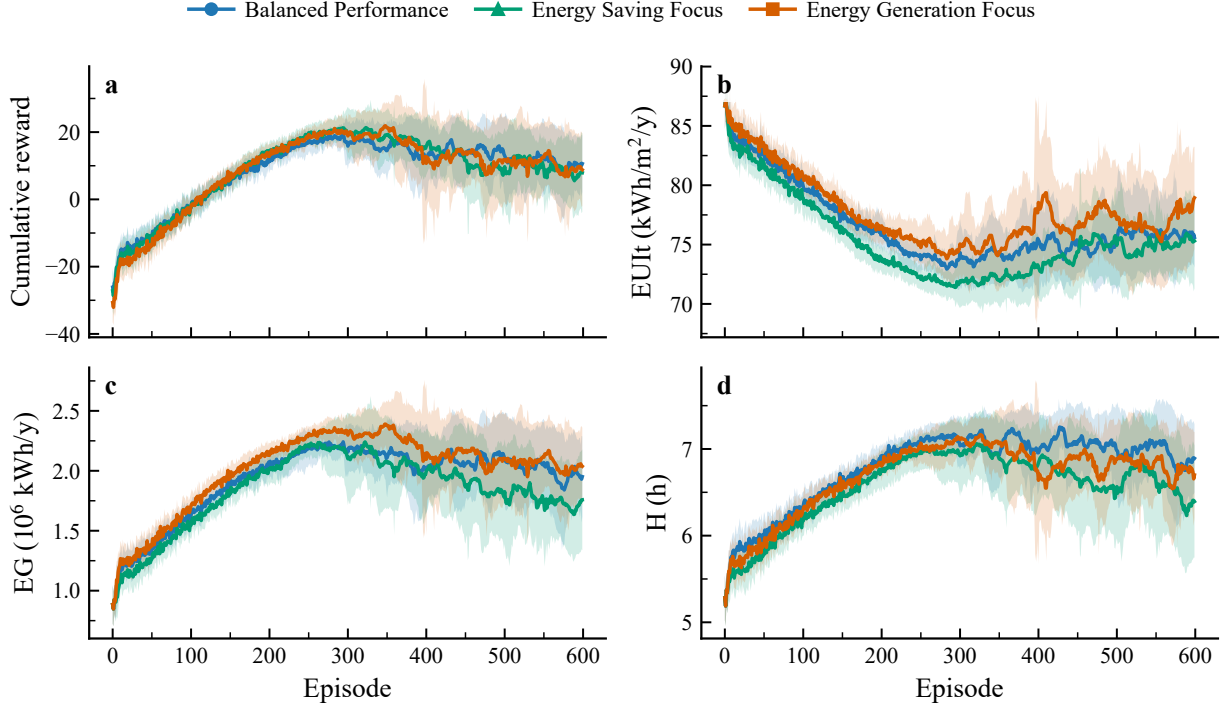


Figure 6: DDPG learning curves during guarded multi-seed training. Solid lines denote mean trajectories across 20 seeds per scenario and shaded regions indicate one standard deviation.

3.3. Multi-objective Optimization Results

Before presenting the comparative results, it is necessary to clarify how the DRL solution clusters were generated. To ensure statistical robustness, each of the three DRL scenarios was trained independently across 20 random seeds. This produces a distribution of high-performance solutions for each agent, reflecting a recurrent high-performing region visited by its learned policy rather than a single stable convergence point. The analysis below first explores the learned tendencies of the three DRL scenarios and then compares them against the NSGA-II benchmark.

3.3.1. Multi-scenario DRL Behavior

The primary strength of the DRL framework lies in its flexibility to generate targeted design solutions. By specifying a design preference a priori through reward-function weights, the framework trains three reward-weighted agents under the same surrogate environment. Figure 7 illustrates the differences among the scenarios. On the selected surrogate, the three DDPG scenarios still occupy overlapping regions, but the scenario-level objective differences remain readable through their morphology summaries.

The most important observation is that the reward-conditioned DRL agents still identify coherent preference-conditioned surrogate regions, but they do not surpass the fair-budget NSGA-II archive on the selected surrogate. As shown in Fig. 7, the three DDPG scenarios form overlapping clusters, whereas NSGA-II reaches better average objective values and stronger Pareto indicators on the same 2000-sample surrogate. This supports the narrower claim that policy learning can identify a coherent region of the surrogate landscape, but it also shows that the tested reward-weighted DDPG formulation is not competitive with fair-budget NSGA-II under the present benchmark setting.

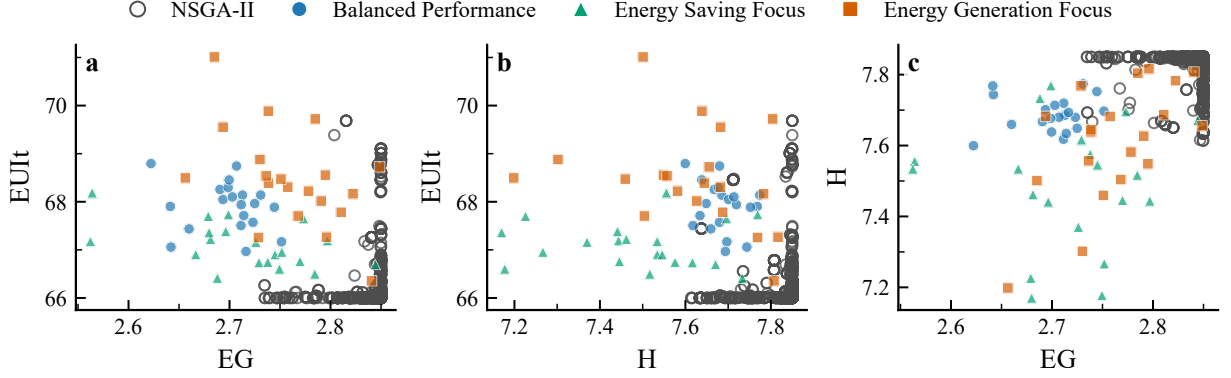


Figure 7: Pairwise objective-space comparison on the selected 2000-sample surrogate. Each point is one candidate solution. The three DDPG scenarios still form overlapping clusters, while the fair-budget NSGA-II archive reaches stronger objective values on the same surrogate.

These objective-space differences become more interpretable in the morphology signatures shown in Fig. 8. The heatmap summarizes relative median UMF levels for each method. On the selected surrogate, the balanced scenario tends toward relatively higher OSR and SVF, the generation-focused scenario shifts toward higher SD and OSLI while retaining high SVF, and the saving-focused scenario combines lower BD with comparatively higher FAR and AF. The contrasts remain moderate and overlapping rather than sharply separated, but they still indicate readable morphology-level fingerprints that vary with the reward specification within the current surrogate workflow.

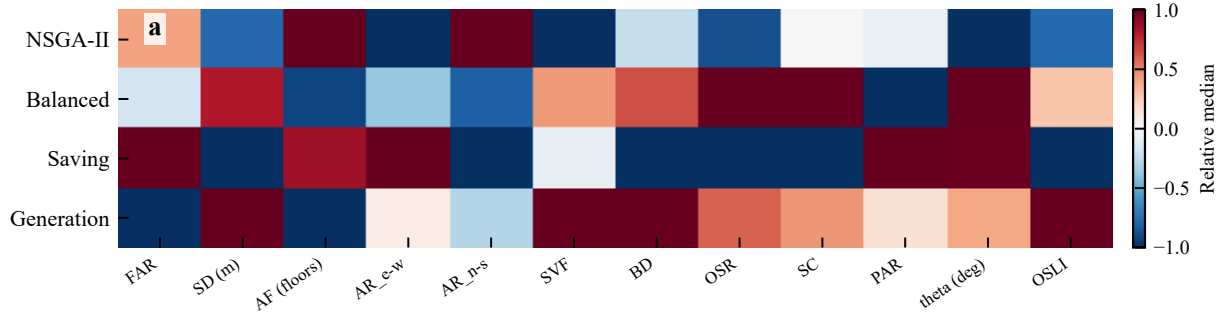


Figure 8: Relative median morphology signatures across methods. Colors encode within-factor contrast across the four method-level median profiles, highlighting which UMFs shift most under the three DDPG reward settings relative to NSGA-II.

3.3.2. Comparison with NSGA-II

After establishing the internal behavior of the DRL agents, we compare them to the NSGA-II benchmark on the selected surrogate from the scale study. Table 6 summarizes this apples-to-apples comparison. On this result set, NSGA-II again attains the strongest Pareto-quality metrics ($HV = 1.331$, $IGD = 0.107$), while the three DDPG scenarios remain weaker. The generation-focused DDPG run is the strongest of the three DDPG variants in Pareto-quality terms ($HV = 1.011$, $IGD = 0.220$), but it still trails NSGA-II. Figure 9 visualizes the same ordering from two complementary angles: scenario-wise objective summaries and preference-aligned utility.

This result leads to a narrower interpretation of the benchmark comparison. Rather than claiming superiority of DRL over NSGA-II, the current evidence supports the conclusion that DDPG does not match fair-budget NSGA-II on the selected surrogate. Under post-hoc preference-aligned utility selection, the balanced, saving-focused, and generation-focused DDPG candidates all remain below the corresponding best NSGA-II candidate, with utility deltas of approximately -0.267 , -0.140 , and -0.046 , respectively. This conclusion should be interpreted together with three additional caveats. First, the independent re-evaluation reported in Appendix B was also carried out for the

2000-sample comparison, but it still relies on the fallback analytic simulator rather than a full physical stack. Second, the DDPG runs continue to exhibit frequent late-stage reward regression despite reaching high-reward regions mid-training. Third, the optimizer ranking remains checkpoint-sensitive, which means benchmark conclusions should be tied explicitly to the selected surrogate artifact rather than generalized across checkpoints.

Detailed linear correlation maps, joint distributions, and nonlinear response diagnostics are reported in the Supplementary Material rather than in the main text. In the present study, these diagnostics are used only as interpretive support for the morphology-level shifts in Fig. 8; they are not treated as primary evidence for the benchmark comparison.

Table 6: Comparison across fair-budget NSGA-II and the three DDPG scenarios on the selected 2000-sample surrogate.

Method	HV	IGD	EUIt	EG (10 ⁶ kWh/y)	H (h)
NSGA-II	1.331	0.107	66.15	2.844	7.836
Balanced DDPG	0.347	0.720	67.90	2.700	7.687
Saving DDPG	0.625	0.549	67.10	2.716	7.487
Generation DDPG	1.011	0.220	68.46	2.763	7.612

Note: The values in this table are computed on the selected 2000-sample surrogate from the scale study. On this surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in both archive-level Pareto metrics and preference-aligned utility.

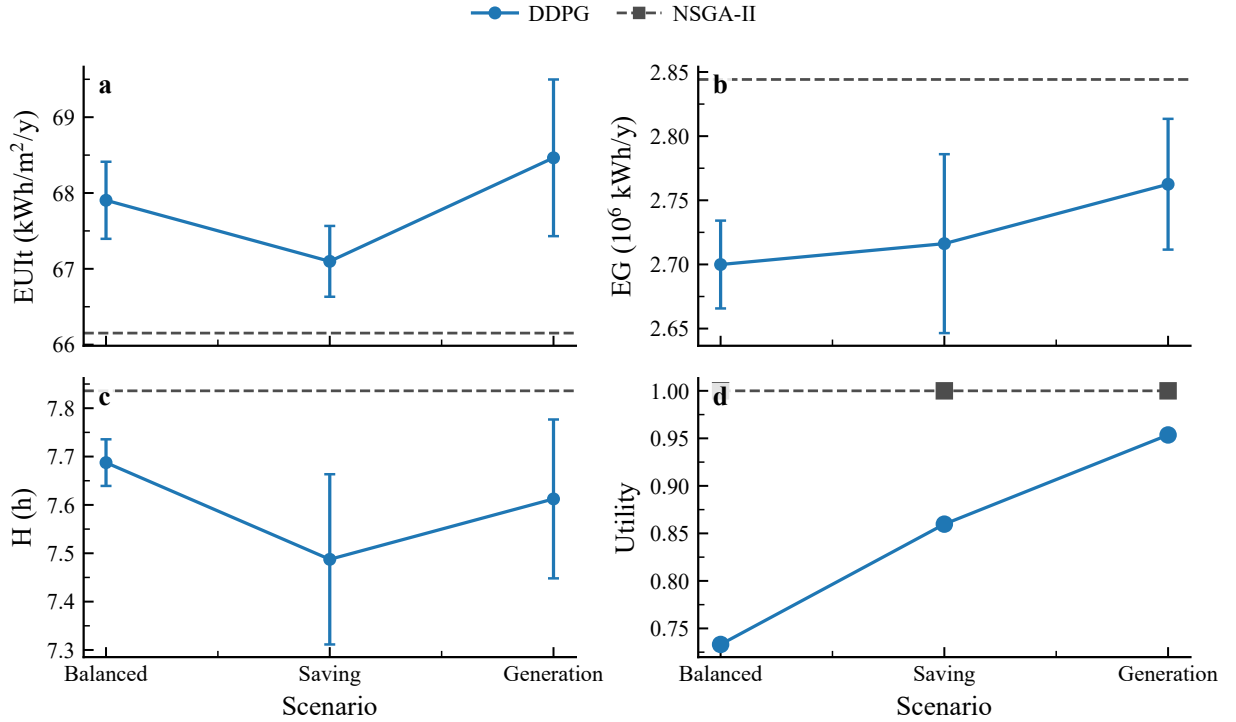


Figure 9: Benchmark summary on the selected 2000-sample surrogate. Panels (a)–(c) compare scenario-wise objective summaries for DDPG, with the NSGA-II mean shown as a dashed reference line. Panel (d) compares preference-aligned utility for DDPG and NSGA-II across the three scenarios.

Figure 10 summarizes the seed-level convergence diagnostics on the selected 2000-sample surrogate. The generation-focused scenario reaches the highest mean best reward, whereas the saving-focused scenario reaches its plateau earliest. However, all three scenarios still show large best-to-final reward gaps and high late-regression fractions, which confirms that the main role of DDPG in the present workflow is region-finding rather than stable monotonic improvement to a final fixed point.

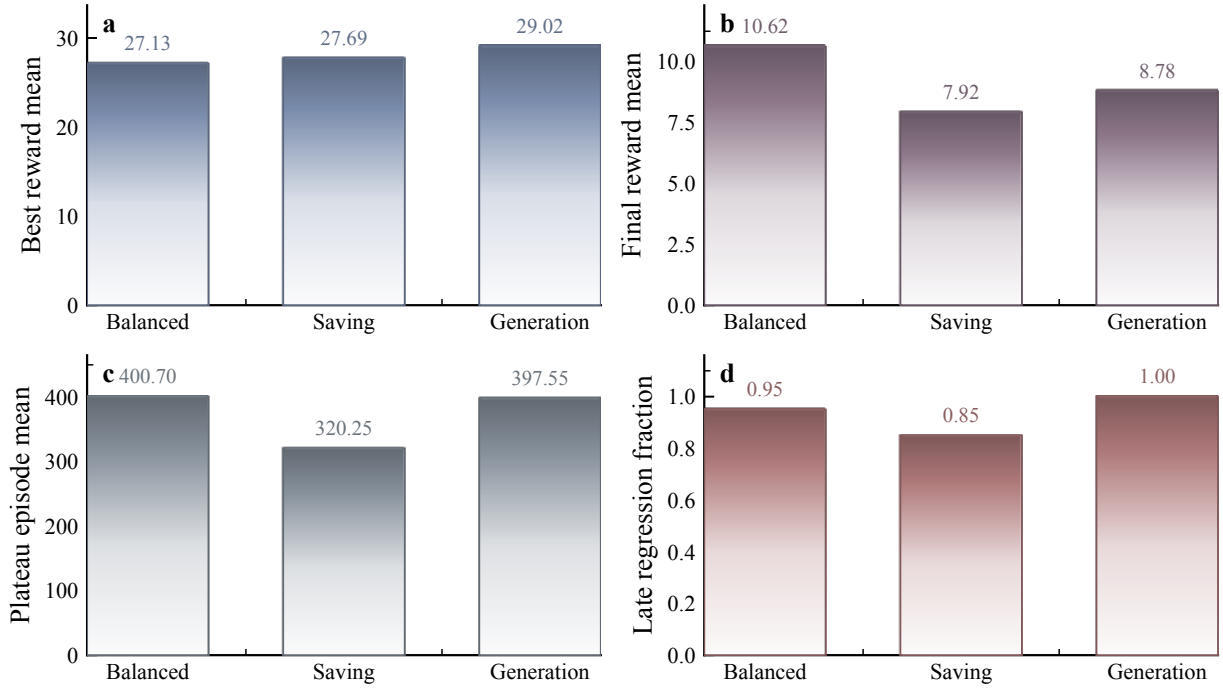


Figure 10: Seed-level convergence diagnostics on the selected 2000-sample surrogate. Panels report the mean best reward, mean final reward, mean plateau episode, and late-regression fraction across 20 seeds per DDPG scenario.

Figure 11 links the optimizer comparison back to the surrogate-selection procedure itself. The regime-winner diagnostics show that surrogate fidelity improves steadily as the dataset grows from 500 to 2000 samples, and that the 2000-sample DNN with standard scaling provides the strongest predictive accuracy among the tested candidates. This figure is included to make the paper’s final optimizer benchmark traceable to an explicit surrogate-selection decision.

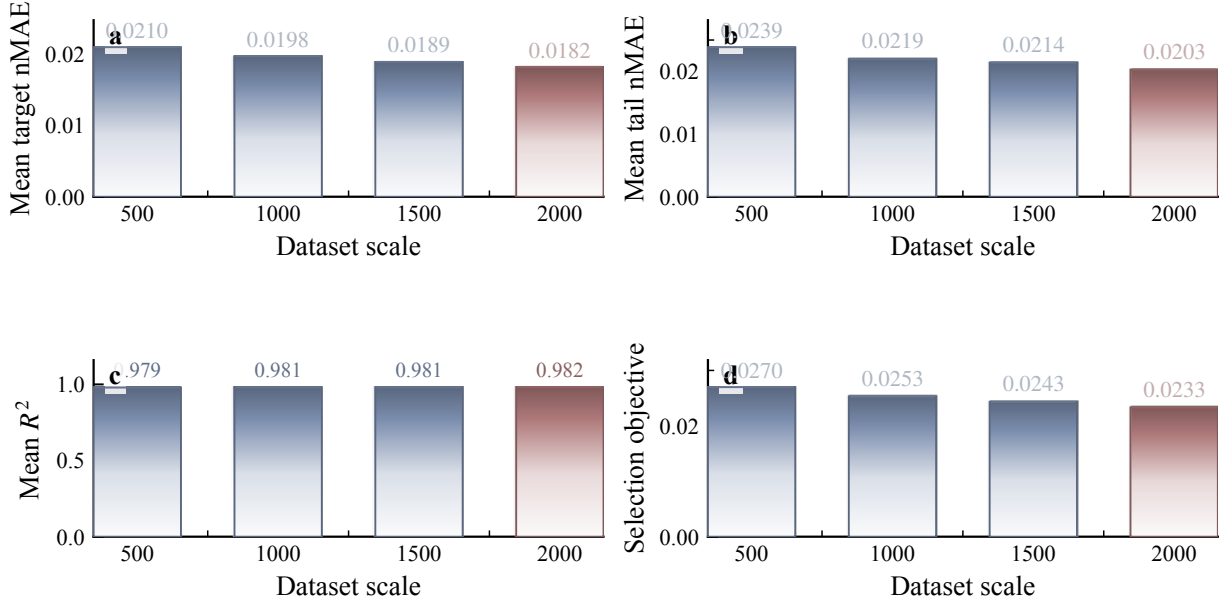


Figure 11: Surrogate scale-study diagnostics across the best regime winner at each dataset scale. Panels report mean target nMAE, mean tail nMAE, mean R^2 , and the selection objective from 500 to 2000 samples, with the 2000-sample winner highlighted as the selected model for optimizer comparison.

4. Discussion

This study develops a DRL-DNN integrated framework for block-scale urban morphology design and examines its comparative performance under a surrogate-selection protocol based on predictive accuracy. The analysis indicates that the framework is most credible as a preference-conditioned design-search tool and as a methodological case study of surrogate-dependent benchmarking, rather than as a superior substitute for Pareto-based evolutionary optimization.

The main methodological finding is that reward-conditioned policy learning can guide the search toward compact high-performing regions on a given surrogate checkpoint, but that advantage remains less stable and less competitive than the NSGA-II baseline. On the selected 2000-sample surrogate, fair-budget NSGA-II is clearly stronger than all three DDPG scenarios in both objective summaries and Pareto-quality metrics, and it also dominates the post-hoc preference-aligned utility comparison. The three DDPG scenarios remain distinguishable but overlapping in objective space, which suggests that the value of DRL in the present setting lies more in region-finding and interpretable morphology conditioning than in producing clearly separated preference optima.

Even so, the multi-scenario design underscores the flexibility of the DRL framework as a decision-support tool. Unlike NSGA-II, which returns a broad non-dominated archive, the proposed workflow allows designers to encode preferences in advance. On the selected surrogate, the resulting differences are clearer in morphology descriptors than in top-line objective means, but they can still be interpreted in terms of density, openness, sky access, and orientation.

The framework also provides interpretable design guidance. The correlation analysis and the additional nonlinear diagnostics indicate that openness-related variables such as OSR and SVF, together with density-related variables such as FAR and BD, play a dominant role in shaping the objective responses. Thus, even though DDPG does not outperform NSGA-II on the selected surrogate, the DRL formulation remains useful for generating policy-conditioned design heuristics within the present case study.

Independent candidate re-evaluation further sharpens this interpretation. On the 2000-sample surrogate, the three DDPG winners do not all drift in the same direction: the balanced candidate improves in EUIt but loses EG and H, whereas the saving-focused and generation-focused candidates drift upward in EUIt and downward in EG after

reevaluation through the independent fallback simulation path. The selected NSGA-II candidate also drifts away from its surrogate optimum, but it remains competitive in aggregate objective value. These results reinforce the need to keep optimizer claims bounded by surrogate reliability rather than by surrogate-space optimum values alone.

Several limitations should be acknowledged. First, the findings are derived from a single case study in one climate zone. Second, even with 2000 simulated cases, the surrogate still approximates a 12-dimensional continuous design space and therefore remains vulnerable in sparse or extreme regions; accordingly, the current Pareto claims should be interpreted as applying to the interpolated surrogate region rather than to unconstrained extrapolation beyond the sampled design domain. Third, the current comparison remains surrogate-based during optimization, and the independent re-evaluation still relies on a fallback analytic simulator rather than a full physical stack or external empirical validation dataset. Fourth, the reevaluation shows that both DDPG and NSGA-II candidates can drift materially relative to their surrogate predictions, so surrogate-space ranking should not be overgeneralized. Fifth, although the current comparison closes the per-seed and seed-count symmetry between DDPG and NSGA-II, broader optimizer families still need to be tested before claiming general policy-learning advantages. Sixth, the present RL formulation is a scalarized static black-box optimization problem recast as policy learning; therefore, alternative continuous optimizers remain important baselines for future work.

5. Conclusions

This study presents a DRL-DNN framework for co-optimizing urban morphology and energy performance in a surrogate-based setting. The evidence suggests that the method should be interpreted as a preference-conditioned policy-learning framework rather than as a universally dominant alternative to Pareto-based search.

Three main conclusions can be drawn. First, the dataset-scale study identifies the 2000-sample DNN with standard scaling as the most accurate surrogate among the tested candidates. Second, when the optimizer comparison is conducted on that surrogate within the paper’s three-scenario DDPG-versus-NSGA-II architecture, fair-budget NSGA-II outperforms all three DDPG scenarios in archive-level objective summaries, Hypervolume, IGD, and preference-aligned utility. Third, the DRL scenarios still yield interpretable design fingerprints, particularly through differences in FAR, BD, OSR, SVF, and OSLI, which can be explained through the correlation and joint-distribution analyses.

Overall, this paper reformulates the optimization problem from a static black-box search into a policy-learning framework that can encode design preferences directly. The resulting workflow is best viewed as a flexible and interpretable design-search tool whose claims must remain bounded by surrogate reliability, benchmark fairness, and the absence of external empirical validation. Future work should expand the dataset, broaden the optimizer baselines, and validate the final solutions through direct simulation and external data.

Appendix A. Additional Case Settings and Surrogate Validation

Appendix A.1. Case-setting details

The base urban block is represented as a $240\text{ m} \times 240\text{ m}$ square subdivided into a 3×3 grid of nine land units. Each land unit measures $80\text{ m} \times 80\text{ m}$. A two-step inward offset is applied to represent the road boundary and building setback, resulting in a nominal buildable area of $60\text{ m} \times 60\text{ m}$ for each unit. One land unit is designated as open space, and the remaining units are populated with residential building prototypes.

To avoid notation drift between the main text and the appendix, the following definitions are used consistently:

- **AF** (average floors) is defined as the total gross floor area divided by the total building footprint area.
- **OSR** (open space ratio) is defined as the non-built-up area of the block divided by the total gross floor area.

Table A.7: Representative building prototypes used in the parametric dataset.

ID	Type	Footprint Area (m ²)	Floors	FAR Range
P-1	Point	2000	1–3	0.31–0.94
P-2	Point	1500	4–12	0.94–2.81
P-3	Point	1000	13–30	2.03–4.69
P-4	Point	1000	13–30	2.03–4.69
S-1	Slab	2000	1–3	0.31–0.94
S-2	Slab	1500	4–12	0.94–2.81
S-3	Slab	1000	13–30	2.03–4.69
C-1	Courtyard	2000	1–3	0.31–0.94
C-2	Courtyard	1500	4–12	0.94–2.81

Appendix A.2. Surrogate validation

The original surrogate workflow began from a 500-sample baseline, but the final optimizer comparison in this revision uses the strict highest-accuracy surrogate obtained from the expanded 2000-sample regime. In the synced artifact bundle, that regime is generated under the documented `fallback_analytic` simulation mode rather than by a direct full-stack EnergyPlus/Radiance rerun. Five-fold cross-validation is used to assess that final selected model. Table A.8 reports the corresponding aggregate validation results.

Table A.8: Five-fold cross-validation summary for the DNN surrogate model.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUI _t	0.983	0.209	± 0.001	± 0.005
EG	0.972	0.026	± 0.002	± 0.001
H	0.992	0.025	± 0.001	± 0.001

To evaluate reliability in optimization-relevant regions, additional error slices were computed over the lower and upper tails of each target distribution. The corresponding mean absolute errors are listed in Table A.9. Table A.10 reports complementary surrogate-fidelity statistics from the cross-validation predictions.

Table A.9: Error diagnostics in extreme regions of the target distributions.

Target	MAE _{all}	MAE _{q_{10}}	MAE _{q_{90}}	$q_{0.10}$	$q_{0.90}$
EUI _t	0.209	0.283	0.194	67.83	73.32
EG	0.026	0.027	0.030	2.132	2.655
H	0.025	0.026	0.029	6.514	7.429

Table A.10: Additional surrogate-fidelity audit on the cross-validation predictions of the selected 2000-sample surrogate.

Target	RMSE	MAE	nRMSE	Tail nMAE	R^2
EUI _t	0.279	0.209	0.0227	0.0194	0.983
EG	0.033	0.026	0.0305	0.0265	0.972
H	0.032	0.025	0.0172	0.0150	0.992

Note: nRMSE is normalized by the observed range of each target over the cross-validation predictions. Tail nMAE is the mean of the lower-tail and upper-tail nMAE diagnostics used for surrogate selection.

Appendix B. Benchmark Robustness and Independent Re-evaluation

Appendix B.1. Training dynamics and benchmark protocol

The revised server-side reruns used for the final accuracy-first comparison were executed under the shared guarded surrogate evaluator to suppress exploitation of extrapolative edge cases. On the strict highest-accuracy 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401 (balanced), 320 (saving), and 398 (generation). Compared with the earlier lower-scale rerun, the balanced scenario becomes visibly smoother and less noisy across seeds, but the saving-focused and generation-focused scenarios do not show a uniform reduction in cross-seed variability across reward and objective trajectories. However, late-stage regression remains common: the mean best-to-final reward gap ratios are about 0.62, 0.71, and 0.71 for the balanced, saving, and generation scenarios, respectively, and the substantial-regression fractions remain high at 95%, 85%, and 100%.

For benchmark comparison, the revised implementation removes the previous benchmark-matching shortcut from the publication-mode NSGA-II path. Publication mode compares DDPG and NSGA-II under a shared surrogate environment, matched per-seed evaluation budget, and matched seed count, and applies a post-hoc preference-aligned utility summary consistently across methods. On the strict highest-accuracy 2000-sample surrogate, NSGA-II remains stronger than all three DDPG scenarios. The corresponding budget accounting and objective-space separation are summarized in Table B.11.

Table B.11: Budget accounting and objective-space separation in the highest-accuracy 2000-sample rerun.

Item	Value 1	Value 2	Interpretation
DDPG per-seed budget	600 episodes	40 steps/episode	24,000 surrogate interactions
DDPG seeds per scenario	20	–	pooled into one scenario archive
NSGA-II seeds in rerun	20	–	pooled into one comparison archive
Distance: balanced vs generation	0.5417	–	normalized objective-space distance of scenario means
Distance: balanced vs saving	0.6809	–	normalized objective-space distance of scenario means
Distance: generation vs saving	0.7619	–	normalized objective-space distance of scenario means
NSGA-II archive spread	0.1642	0.1968	mean and std of distance to normalized archive center

Note: The 2000-sample highest-accuracy rerun matches both the per-seed interaction budget and the number of seeds between DDPG and NSGA-II. The DDPG scenarios are more separated than in the earlier matched-checkpoint run, but NSGA-II still dominates the preference-aligned utility comparison.

To complement the linear correlation results in the main text, Fig. B.12 reports representative surrogate-based nonlinear response profiles for $\text{OSR} \rightarrow \text{EUI}$, $\text{FAR} \rightarrow \text{EG}$, $\text{SVF} \rightarrow \text{H}$, and $\theta \rightarrow \text{H}$.

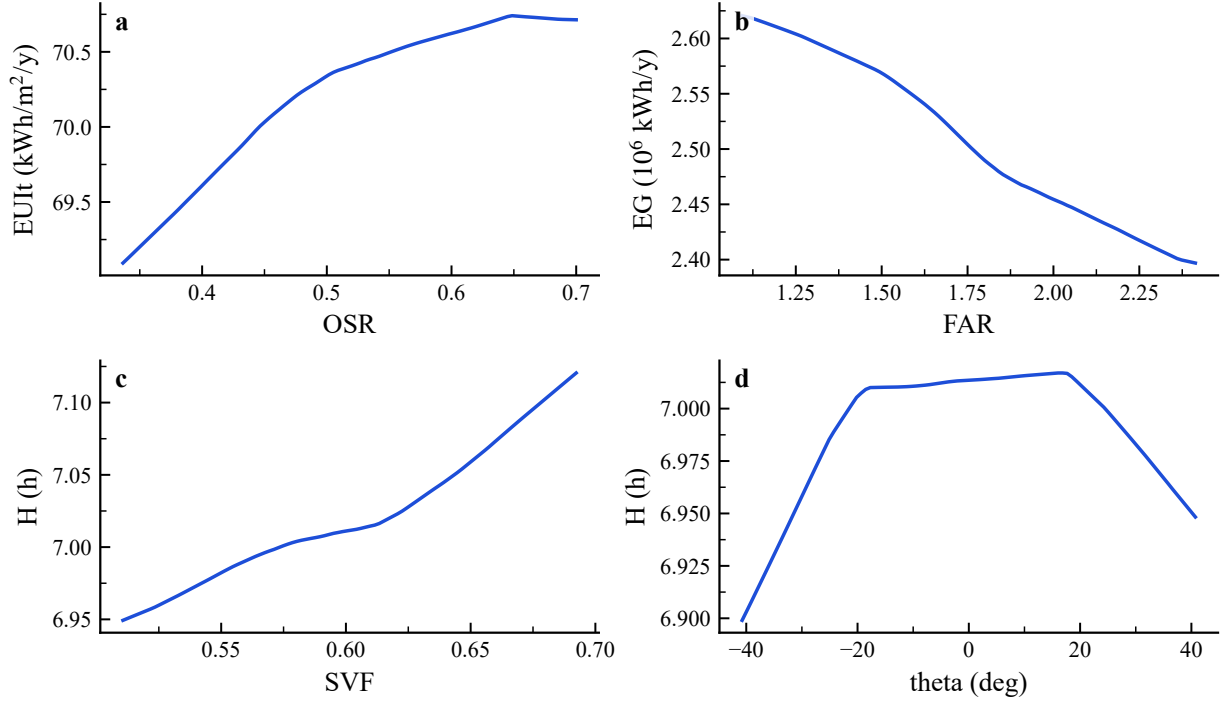


Figure B.12: Surrogate-based nonlinear response profiles for selected UMF-target pairs: (a) OSR→EUI_t, (b) FAR→EG, (c) SVF→H, and (d) $\theta \rightarrow H$.

Appendix B.2. Independent re-evaluation of selected candidates

To reduce over-reliance on surrogate predictions, the workflow re-evaluates preference-aligned candidate solutions using an independent simulation call. In the current local environment, this re-evaluation still uses the fallback analytic simulator, but the code path is structured so that a full physical-stack simulation can replace it when server-side results become available. Table B.12 reports the latest available reevaluation results for the strict highest-accuracy 2000-sample comparison. The updated evidence shows heterogeneous drift: the balanced DDPG winner improves in EUI_t but loses EG and H, the saving-focused and generation-focused DDPG winners drift upward in EUI_t and downward in EG, and the selected NSGA-II candidate also moves away from its surrogate prediction. This narrows the earlier asymmetry between DDPG and NSGA-II, but it still indicates that surrogate-space ranking should not be treated as a direct substitute for independent validation.

Table B.12: Independent re-evaluation of selected candidates from the highest-accuracy 2000-sample surrogate rerun. Surrogate and deterministic re-evaluation values are reported for the current preference-aligned selections.

Method	Scenario	EUI _{t,sur}	EUI _{t,re}	EG _{sur}	EG _{re}	H _{sur}	H _{re}
DDPG	Balanced	67.168	66.000	2.751	2.689	7.697	7.374
NSGA-II	NSGA-II (seed 20260310)	66.000	67.730	2.850	2.712	7.850	7.510
DDPG	Saving	66.698	67.473	2.844	2.769	7.671	7.611
DDPG	Generation	66.356	67.836	2.841	2.735	7.807	7.530

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